

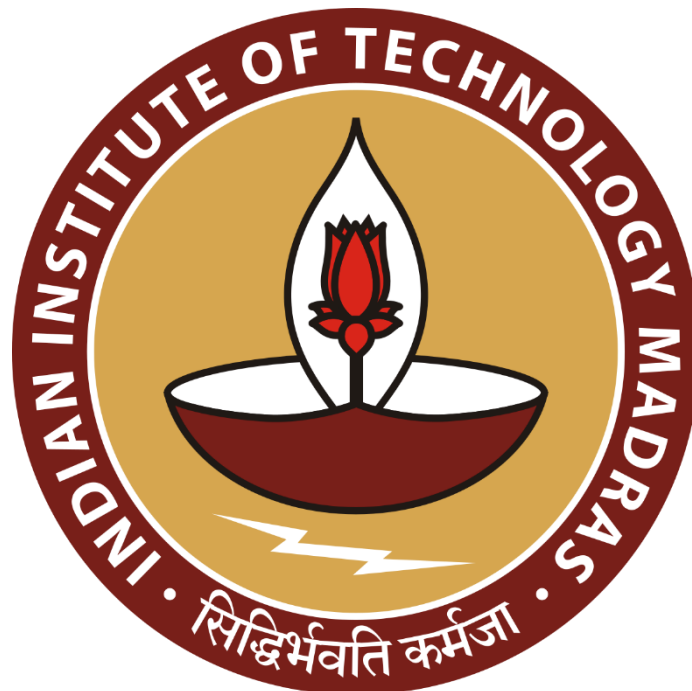
Analytical Study of Member Lifecycle and Revenue Leakage at Muscle and Fitness 2

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “Analytical Study of Member Lifecycle and Revenue Leakage at Muscle & Fitness Gym 2”. I extend my appreciation to **Muscle and Fitness 2**, for providing the necessary resources that enabled to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate:

Barun Sharma.

Name: Barun Sharma

Date: February 21, 2026

1 Executive Summary

The project focuses on studying and providing data-driven recommendations for the challenges faced by “**Muscle and Fitness 2**,” a local fitness center operating for the past 5–6 years in Agartala, Tripura. The business functions in a B2C model and offers monthly, quarterly, and yearly membership plans to members from the nearby community who seek affordable fitness and health services.

The major challenges faced by Muscle and Fitness 2 arise from the absence of a structured data management system. All member and payment records are maintained manually in physical registers. Due to this, the owner is unable to clearly track monthly revenue trends, identify regular and irregular fee payments, or understand how many members continue or drop out over time. This lack of visibility leads to confusion in financial tracking and limits effective decision-making.

To address these challenges, this project explores six months of membership fee data maintained by the gym. The dataset includes details such as payment dates, member IDs, amounts paid, and modes of payment. The analysis will focus on understanding monthly revenue flow and examining membership continuity and payment behaviour.

By transforming raw register data into organized and meaningful insights, the project aims to help the owner gain clarity on financial health, reduce revenue leakage, improve member retention, and adopt a more reliable, data-driven approach to managing gym operations.

2 Organisation Background

Business Name: Muscle and Fitness 2

Address: Lane No. 5, Dhaleswar, Agartala, Tripura 799004

Owner's Name: Mr. Rupak Biswas

Muscle and Fitness 2 is a local fitness center that has been serving the community for the past **six to seven years**, promoting fitness and a healthy lifestyle. The gym offers a variety of membership plans, including **monthly, quarterly, and yearly packages**, catering to individuals of all age groups and fitness levels. In addition to general membership, the gym also provides **personal training programs** for members seeking customized guidance.

The gym employs **two professional trainers**, who work in **morning and evening shifts**. The morning trainer primarily assists working professionals and early-morning fitness enthusiasts, while the evening trainer focuses on students and residents who prefer post-work sessions. This structured shift system ensures continuous member support and effective utilization of gym resources throughout the day.

Located in a moderately populated residential area, the gym primarily serves **students, working individuals, and local residents** who seek affordable yet effective fitness solutions. It offers essential workout equipment and provides personal guidance in **strength training, weight management, and overall fitness improvement**.

The owner, **Mr. Rupak Biswas**, personally handles all **member registrations, fee collections, and payment records**. However, all financial and attendance information is maintained manually in physical registers. This traditional old approach makes it difficult to track performance, monitor revenue trends, and evaluate member retention. With the rise of modern fitness centers adopting digital management systems, the owner now recognizes the need to implement a **data-driven management approach** to improve operational efficiency, transparency, and business growth.

3 Problem Statement

3.1 Inability to Track Monthly Revenue and Financial Performance

The gym lacks a structured system to summarize and analyse monthly fee collections, making it difficult to assess revenue trends, profitability, and financial growth over time.

3.2 Absence of Member Retention and Continuity Analysis

There is no mechanism to track membership renewals or drop-offs, which prevents the gym from identifying loyal members, measuring churn, and improving long-term member retention strategies.

3.3 Lack of Proper System to Monitor Member Payments

The absence of an automated payment tracking system results in expired memberships going unnoticed, leading to revenue leakage, manual verification efforts, and errors in payment monitoring.

4 Background of the Problem

The issues faced by the ‘**Muscle and Fitness 2**’ come from the very nature of their manual record-keeping. Firstly, figuring out how much money comes in each month and whether the gym is growing or not is a problem due to the lack of a proper system. All transactions are

written down in a physical register, which makes it very difficult to get a clear financial picture without manually going through every page. This makes planning for expenses and setting future income goals a challenging task.

Secondly, the owner is struggling with understanding loyal members. Without a system to track memberships properly, it's hard to know who is a long-term member and who quits just after their first month. This lack of information makes it difficult to create strategies to keep members coming back or to reward those who are loyal. So, it is necessary to have a solid system to build a stable customer base.

In summary, the root cause of the problems is from the lack of maintenance of transaction and membership data in the gym, combined with the absence of an automated tracking system. Addressing these challenges becomes vital for bringing financial clarity and smooth operations to the business.

5 Problem Solving Approach

5a. Details about the methods used with Justification:

Given the nature of the problems faced by the gym, a comprehensive approach involving both quantitative and qualitative methods is required to develop an effective solution.

- **Quantitative Methods:**

- **Time-Series Analysis:** The gym's financial and membership data is inherently time-dependent. This method will be used to analyse monthly revenue, track membership growth or decline over time, and identify seasonal patterns (e.g., a surge in new members in January).
- **Statistical Computation:** Key Performance Indicators (KPIs) will be calculated to assess the health of the business. This includes computing the **monthly revenue**, **member churn rate**, **retention rate**, and **average revenue per member (ARPM)** to provide clear, data-driven insights into the gym's performance.

- **Qualitative Methods:**

- **Conversation:** I will engage with the gym owner to gather qualitative insights into their current manual processes, specific frustrations with payment tracking, and the desired outcomes from a new system. This ensures the final solution meets their practical needs.
- **Benchmarking:** I plan to research standard practices at similar local gyms to establish benchmarks for member retention and engagement strategies. This will help in setting realistic goals and identifying effective industry practices.

5b. Details about the intended data collection with Justification:

To address the core problems, two primary categories of data need to be collected from the gym's existing physical records.

- **Member Payment Data:** This involves collecting detailed records of all member payments.
 - **Variables:** Member ID, Payment Date, Amount Paid, Membership Type (e.g., monthly, quarterly, annual), and Payment Method.
- **Membership Status Data:** This involves gathering information on the lifecycle of each member.

Variables: Member ID, Membership Start Date, Membership Expiry Date, and Renewal Status.

- **Justification:**

This dataset is essential for addressing all three problem statements. The **Payment Data** directly enables accurate monthly revenue calculation (**Problem 3.1**) and provides the foundation for a system to track due payments (**Problem 3.3**).

The **Membership Status Data** is crucial for analysing member retention and churn (**Problem 3.2**) by tracking how many members renew their plans upon expiry.

Together, these variables provide a 360-degree view of the gym's financial and operational health.

5c. Analysis Tools and Justification:

- **Analysis Tool:**

Microsoft Excel: This project will be executed entirely within Microsoft Excel, leveraging its comprehensive suite of tools including data tables, built-in formulas (SUMIFS, COUNTIFS, IF), PivotTables, Conditional Formatting, and charting capabilities.

- **Justification:**

Sufficiency and Power: For the scale of this gym's operations, Excel is a powerful and sufficient tool. It can effectively manage member lists, log all financial transactions, and perform the necessary calculations to solve all three problem statements without the need for more complex software.

Accessibility and Cost-Effectiveness: Excel is a readily available tool that requires no additional software investment. Its user-friendly interface makes it

the most practical choice for a small business owner transitioning from a manual, paper-based system for the first time.

Direct Problem-Solving Capability:

- **Revenue Analysis:** PivotTables and charts will provide immediate and clear visualizations of monthly income and growth trends.
- **Retention and Churn:** Member data can be easily manipulated to calculate key metrics like retention and churn rates using straightforward formulas.
- **Payment Tracking:** A combination of date functions (TODAY ()) and Conditional Formatting provide a simple, visual, and effective system for flagging expired memberships daily.

6 Expected Timeline

6.1 Work Breakdown Structure:

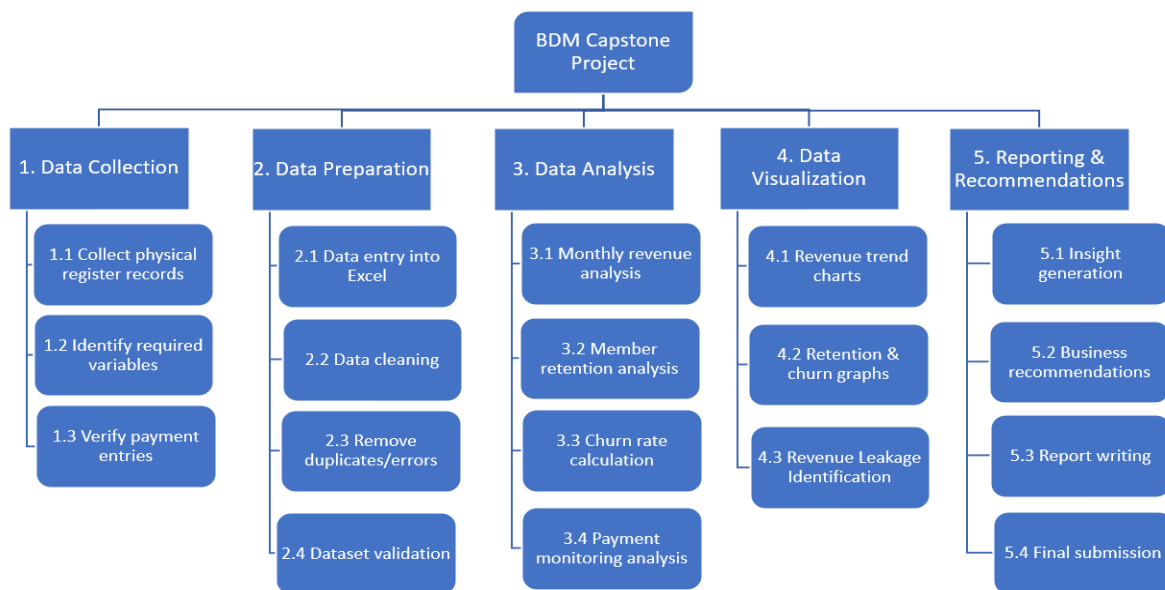


Figure 1: Work Breakdown Structure (WBS) for BDM Capstone Project

6.2 Gantt chart

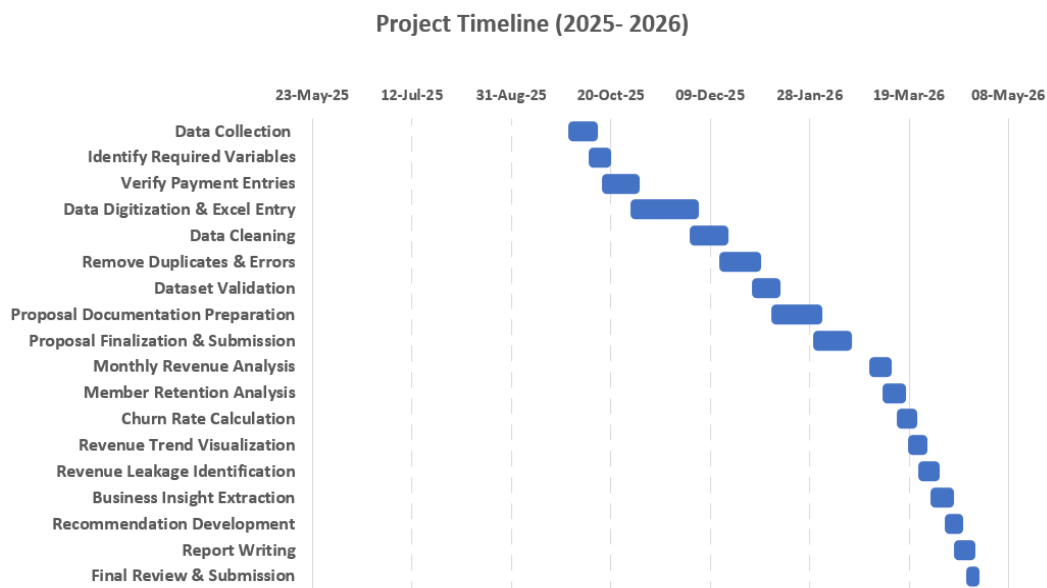


Figure 2: Project Timeline (Gantt chart) for BDM Capstone Project

7 Expected Outcome

The primary outcome of this project is to transition **Muscle and Fitness 2 Gym** from a manual, intuition-based management style to a **data-driven operational model**. By implementing structured profit analysis, the owner will gain the ability to compare total fee collections against fixed monthly expenses—such as rent, electricity, and salaries—to determine the gym's true net profitability for the first time.

Furthermore, the project will deliver a comprehensive **Financial Health Report** that tracks monthly revenue trends and overall business growth, allowing for better long-term planning. A critical operational outcome will be the elimination of revenue leakage; by developing a tracking system in Excel that flags expired memberships via conditional formatting, the owner can immediately identify members who have not paid, effectively stopping money from being lost.

Finally, the analysis will provide a "360-degree view" of the member lifecycle. By grouping members into categories like **"Loyal," "New," or "Inactive,"** the business owner can identify churn patterns and implement targeted retention strategies to keep members engaged

longer, ultimately stabilizing the gym's customer base and increasing long-term sustainability.